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Efficient semantic network construction with application to PubMed search ☆

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ABSTRACT

Exploring PubMed to find relevant information is challenging and time-consuming because PubMed typically returns a long list of articles as a result of query. Semantic network helps users to explore a large document collection and to capture key concepts and relationships among the concepts. The semantic network also serves to broaden the user's knowledge and extend query keyword by detecting and visualizing new related concepts or relations hidden in the retrieved documents. The problem of existing semantic network techniques is that they typically produce many redundant relationships, which prevents users from quickly capturing the underlying relationships among concepts. This paper develops an online PubMed search system, which displays semantic networks having no redundant relationships in real-time as a result of query. To do so, we propose an efficient semantic network construction algorithm, which prevents producing redundant relationships during the network construction. Our extensive experiments on actual PubMed data show that the proposed method (COMPACT) is significantly faster than the method removing redundant relationships afterward. Our method is implemented and integrated into a relevance-feedback PubMed search engine, called RefMed, "http://dm.postech.ac.kr/refmed".

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1. Introduction

PubMed-Medline is a biomedical and life science literature repository that contains 18 million articles and keeps growing. It is one of the most important knowledge sources for medical doctors and biomedical researchers. PubMed supports an exact matching search using attributes including title, author, abstract, journal, MeSH terms.[1] However, exploring PubMed to find relevant information is challenging and time-consuming because PubMed typically returns a long list of articles as a result of query.

Semantic network is an approach that improves the quality of search results on large corpus such as PubMed [1]. It presents underlying relationship among concepts in the corpus so that users deeply understand the keyword and help extend the query keyword. Typical semantic network techniques analyze the relationship between concepts by considering co-occurrence of concepts [2–7]. For example, one of fundamental relationships among concepts is superconcept–subconcept relationship. If the document set appearing a certain concept x is included in another document set appearing another concept y, we call y is a superconcept of x, and x is a subconcept of y.

The problem of existing semantic network generating techniques is that, there are so many superconcept–subconcept relationships, and many of them are redundant thus do not provide additional information. Instead, they make it hard to discover useful underlying relationships. For example, in Fig. 1, there are six superconcept–subconcept relationships. However, among them, three relationships (i.e., relationship ④, ⑤, ⑥) do not provide additional information to user. Identifying and removing such redundant relationships requires high computational cost. To deliver information clearly, we need to remove redundant relationship, however, removing such redundant relationship requires high complexity.

In this paper, our goal is to propose a new method which efficiently constructs semantic network having no redundant relationships. Our key observation is that preventing redundant relationships during construction is more efficient than removing redundant relationship after construction. Based on the observation, we first define *minimal semantic network* and *compact*

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[1] http://www.ncbi.nlm.nih.gov/mesh.

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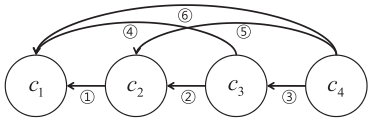


Fig. 1. Example of redundant relationships. Nodes are concepts and edges indicate superconcept–subconcept relationships.

semantic network and develop algorithm to construct the semantic network. In addition, we implement proposed method, COMPACT, in a real-world application called RefMed. RefMed is a a feedback-based interactive search system for PubMed. We add new functionality to RefMed and evaluate the COMPACT on the basis of RefMed query.

Remainder of this paper is organized as follows. The related work is summarized in Section 2. Section 4 presents the proposing method. Section 5 introduces an application which exploits the proposing method in document retrieval task on PubMed. Experimental results are presented in Section 6.

2. Related work

2.1. Various efforts to enhance the search quality on PubMed

To improve the PubMed search quality, researchers have studied querying methodologies for PubMed, such as how to use controlled vocabulary, MeSH terms, or background knowledge to formulate proper PubMed queries [8,9]. Re-organizing the search results using ontologies or clustering techniques has been explored to provide better presentation of the results to the users [10]. These techniques are useful when users do not know what he want exactly, or he need to find new domain. Text mining researchers have also tried to compute the global importance of articles using the citation information and have applied it to rank the results as done in Google [11,12,10,13]. These techniques can enhance the overall document retrieval performance, and can detect articles which does not include the query keywords directly. Researchers have also applied machine learning techniques to find relevant articles by ranking the articles according to the learned relevance function [14,15]. These kinds of results help to rank more relevant articles on the top of result article list. Relevance feedback, a well established technique in IR to improve retrieval performance [16,17], has been applied on PubMed (e.g., MiSearch [18] and RefMed [19]). This approach is especially useful when user already has specific interest, but suffers from the amount of result articles, which is usually over 10,000.

2.2. Semantic network

A semantic network is a way of representing knowledge or underlying structure of concepts by using graphical notations [1,20–26]. In addition, according to semantic network, they usually provide more than structure of knowledge. For example, WordNet is one of the well-known semantic network [20]. It is a lexical database which annotates various relationships between English words. Among various relationships, one of important relationship in WordNet is a *synset* which is groups synonyms. Also, it provides a short description on each *synset*, and maintains different semantic relationship among the *synsets*. Similarly, various kinds of semantic networks is proposed [21,22,26,25].

However, most semantic networks have two severe problems. These semantic networks are manually annotated and maintained by domain experts. Thus, it is expensive to annotate and maintain the semantic network. Human-annotated semantic networks may be accurate but less flexible and inefficient to catch new concepts or topics rapidly evolved. Thus, recent researches adapts a semantic network dynamic construction approach to resolve this problem [25,26].

Based on various semantic networks, several efforts to improve search quality have been proposed [23–26]. Among them, [26] proposed a automatic semantic network generating technique. The key idea of the paper is to analyze the structure of semantic network by recursively generating query by using concepts from retrieved documents. In addition, they adapts entropy from information theory, and process further analysis based on content and meta information for the search results. However, since their semantic network is generated from repetitive querying process, it either incurs much time cost to be serviced in online time or requires offline processing step, which dilutes the advantage of automatic semantic network generation; a rapid responds to newly added concept.

2.3. Semantic networks in BioMedical databases

Inspired from the success of semantic network on information retrieval society, a variety of biomedical or biological data are stored as semantic network. For example, MeSHs (Medical Subject Headings), GO (Gene Ontology) [27], and UniProt (Universal Protein) [28] are semantic networks for Biomedical terms. These semantic networks suffers from the same problem of general semantic networks; it requires much cost to maintain and update the semantic networks and less flexible to newly added concepts. For example, MeSH is updated once a year thus slower to adopt emerging concepts. In this paper, we aim to develop a system that dynamically builds semantic network from PubMed database.

3. Preliminary: elements of semantic network

As mentioned in [26], there are various types of semantic networks as follows. Among them, we focused on assertional networks, which are designed to assert proposition. Basic structure and construction process of semantic network is introduced in [4,1]. The semantic network consists of concepts and relationships among concepts. Two concepts may be related to each other as related set or superconcept–subconcept in the network.

3.1. Concept

A concept means considerable terms from a document collection. A term x_i becomes considerable if (1) it occurs more than a certain threshold in the search results and (2) the fraction of the term's probability in the search results over the total document collection is higher than a threshold [4]. For example, a term becomes a concept when it satisfies the following two conditions.

$$p_D(x_i) = \frac{\#\text{ of occurrence of term } x_i \text{ in D}}{\#\text{ of documents in D}} > \theta_{c_1} \tag{1}$$

and

$$\frac{p_D(x_i)}{p_T(x_i)} > \theta_{c_2} \tag{2}$$

where D and T are the retrieved documents and the total document collection, respectively. In particular, we set $\theta_{c_1} = 5\%$ for our system. For example, when we generate a query using keyword "mrsa" in our application, term "resistance" becomes one of concepts, because resistance occurs in 10,484 articles among 13,114 thus p_D(" resistance") > 0.05. On the contrary, common terms such as "the" cannot

be a concept eventhough they appears more than threshold θ_{c_1}. It is because the term also commonly appears in other articles thus did not satisfy the second condition. (In our system, we set $\theta_{c_2} = 5$.)

3.2. Superconcept–subconcept relationship

The superconcept–subconcept relationship is determined in a similar way. If the document set of x_i is almost included in that of x_j but not in the other way, x_j is a superconcept of x_i.

$$p_D(x_j|x_i)\left(=\frac{p_D(x_i,x_j)}{p_D(x_i)}\right) > \theta_p \quad \text{and} \quad p_D(x_i|x_j) < \theta_p \tag{3}$$

where

$$p_D(x_i,x_j) = \frac{\#\text{ of co}-\text{occurrence of } x_i,x_j \text{ in D}}{\#\text{ of documents in D}} \tag{4}$$

Note that when x_j is a superconcept of x_i then $|x_j| > |x_i|$ where $|x_i|$ is the size of documents showing x_i. It is clear because $\theta_p * P_D(x_j) > P_D(x_i, x_j) > \theta_p * P_D(x_i)$. In particular, we set $\theta_p = 0.6$. For example, when we generate a query using keyword "mrsa", term "chromosome" becomes a subconcept of "strain" because 60% of articles including chromosome also include term "strain".

3.3. Importance of concepts

Supppose a term appears 95% of the result documents. Then, it is likely highly related to the query terms but the term is also likely too general and not very surprising (expected by the user). Adding the term in the query may not help reducing the search results, as it does not provide additional information. On the other hand, when the term is too rare in the results, it is less likely related to the query and adding the term in the query could reduce the results more than necessary. Based on the intuition, we use the entropy to measure the importance of the concepts [29,30].

$$G(x_i) = -p_D(x_i)\log(p_D(x_i)) \tag{5}$$

If a term appears too much or too rarely, then its entropy decreases. We will qualitatively evaluate this measure in Section 6.

3.4. Related term

Relationship *related* is also defined based on the co-occurrence ratio of two concepts. Suppose a term x_j appears in most of the documents having x_i and vice versa. Then, it indicates that there is a strong relationship between x_i and x_j. This is called *related* relationship. For example, concepts x_i and x_j become *related* concept to each other if they satisfy following conditions.

$$p_D(x_i|x_j) > \theta_s \quad \text{and} \quad p_D(x_j|x_i) > \theta_s \tag{6}$$

4. Efficient semantic network construction

In this section, we proposed a efficient semantic network construction method called COMPACT. More specifically, we (1) define redundant relationship in more formal way, (2) introduce post-processing method for removing redundant relationships, (3) introduce new concepts, compact network and minimal network, (4) and propose a method which prevents redundant relationship during network construction.

4.1. Redundant relationship

Detecting the superconcept–subconcept relationships between every pair of concepts often results in generating many redundant relationships that are unnecessary thus to be removed. For example, consider Fig. 1 as the resulting relations of four concepts. Each node represents a concept and each directed edge represents a superconcept–subconcept relationship between two connected nodes. A directed edge from c_j to c_i means c_i is a subconcept of c_j. In Fig. 1, c_1 is a subconcept of c_2, c_2 is a subconcept of c_3, and c_3 is a subconcept of c_4. While detecting the superconcept–subconcept relationships, c_1 will highly likely to be determined as a subconcept of c_3. But, this relationship does not deliver any additional information, thus is redundant, since the relationship can be induced from the graph having c_1 as a subconcept of c_2 and c_2 as a subconcept of c_3. Similarly, the relationships between c_1 and c_4, and c_2 and c_4 are redundant. Based on this observation, we formally define the redundant relationship as follows.

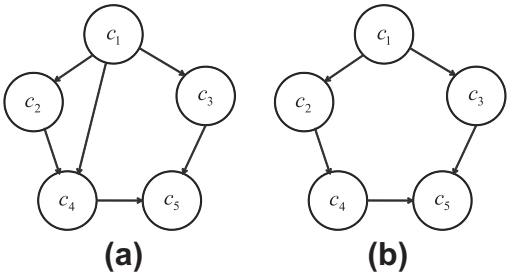


Fig. 2. Example of redundant and non-redundant relationships.

Definition 1 (*Redundant relationship*). Let $c_i \to c_j$ denotes c_i is a superconcept of c_j. A relationship $c_i \to c_j$ is redundant if there is another path $c_i \to c_k \to \cdots \to c_j$ in the semantic network.

As another example, consider Fig. 2a. The superconcept–subconcept relationship from c_1 to c_4 is redundant because there exists another path $c_1 \to c_2 \to c_4$. Note that, our definition does not infer that a path is always redundant if there is another path. For example, although there exist two paths from c_1 to c_5 (i.e., $c_1 \to c_2 \to c_4 \to c_5$ and $c_1 \to c_3 \to c_5$), there is no more redundant relationship in Fig. 2b. It is because any single additional removal of edge cannot be inferred by using other paths.

4.2. Removing redundant relationship

The redundant relationships make the graph complex without providing additional information thus need to be removed. The redundant relationships can be removed by the post-processing after constructing the semantic network, or excluded in advance during construction of semantic network. In our time-critical application, it is important to efficiently remove them. Following subsection presents two algorithms for removing the redundant relationships.

4.2.1. Removing redundant relationships as post-processing

We can detect and remove the redundant relationships after fully constructing the semantic network with redundant relationship. Let $R = \{(c_i, c_j, Type)\}$ be the set of extracted relationships and each relationship is expressed by a triplet $(c_i, c_j, Type)$ where c_i and c_j are concepts and "Type" is the relationship type between c_i and c_j. For example, a triplet $(c_i, c_j, Superconcept)$ indicates "c_i is a *superconcept* of c_j". The post-processing method first extracts R from the document set, and for each triplet $(c_i, c_j, Superconcept) \in R$, if there is another path from c_i to c_j, it removes the triplet from R. The post-processing algorithm is described in Algorithm 1.

Algorithm 1. The post-processing method for removing redundant relationships

Input: a set of concepts C, a document set D, the thresholds θ.
Output: a set of tripets $R = \{(c_i, c_j, Type)\}$.

```
Set R := ∅
// Detect relationships
for i = 1 ... n do
    for j = 1 ... n do
      if p(c_i|c_j) > θ_p and p(c_j| c_i) < θ_p then
        R:=R ∪ (c_i, c_j, Superconcept)
      end if
    end for
end for
// Remove redundant relationships
for each (c_i, c_j, Type) ∈ R do
  if Type = = Superconcept and there is another path from
c_i to c_j
      R:=R − (c_i, c_j, Superconcept)
    end if
end for
return R
```

Note that the complexity of the algorithm is quadratic to the number of triplets in $R(O(|R|^2))$ because, for each triplet in R, examining the possibility of indirect paths takes an additional scan of R. At the worst case, the number of triplets is proportional to the pairs of concepts (i.e., $|R| = O(n^2)$ where n is the number of concepts). Then, the complexity in terms of the number of concepts becomes $O(n^4)$. The post-processing method is too expensive to run in our applications where the semantic network must be constructed in real-time as a result of query. In the following section, we present a novel algorithm for efficiently constructing the "compact" semantic network which excludes redundant relationships during construction of semantic network.

4.2.2. Compact semantic network generation

This section proposes a novel algorithm for generating the *compact semantic network* that has no redundant relationships and thus does not require removing redundant relationships afterward. The algorithm is run in two phases – (1) building *minimal semantic network* and (2) expanding the *minimal semantic network* to the *compact semantic network*.

4.2.2.1. Phase 1. Building the minimal semantic network. We first define two new terminologies – *closest superconcept* and *minimal semantic network*. *Closest superconcept* is defined to be a superconcept that have smallest supporting document size:

Definition 2 (*Closest superconcept*). Let $\mathscr{C}_x = \{c_1 \dots c_k\}$ be a set of superconcepts of c_x. Among them, the smallest concept c_i (i.e., $\arg\min_{c_i} |c_i|$) is the closest superconcept of c_x.

The following Lemma shows that the relationship between a concept c_x and its closest superconcept is always *not redundant*.

Lemma 1. *For any concept c_x in a semantic network $\mathscr{G}$, a relationship between c_x and its closest superconcept is not redundant.*

Proof. We prove this by contradiction. Let c_i be the closest superconcept of c_j (i.e., $c_i \to c_j$). In order for $c_i \to c_j$ to be redundant, there must be a sequence $K = c_1 \to \cdots \to c_n$ connecting c_i and c_j (i.e., $c_i \to K \to c_j$). Then, one of the concept c_k in K is a superconcept of c_j and a subconcept of c_i. That is, $|c_k| > |c_j|$ and $|c_k| < |c_i|$ from the definition of the superconcept–subconcept relationship. Thus, c_i cannot be the closest superconcept of c_j. Therefore, if c_i is the closest superconcept of c_j, $c_i \to c_j$ is not redundant. □

Minimal semantic network is defined to be the semantic network in which the nodes have only the closest superconcepts:

Definition 3 (*Minimal semantic network*). Let graph $\mathscr{G} = (\mathscr{C}, \mathscr{R})$ be a semantic network where $\mathscr{C}$ is a set of concepts and $\mathscr{R}$ is a set of triplets $(c_i, c_j, Type)$. A semantic network $\mathscr{G}'(\subset \mathscr{G}) = (\mathscr{C}, \mathscr{R}'(\subset \mathscr{R}))$ is a minimal semantic network if, for all the superconcept–subconcept relationship $R_k = (c_i, c_j, Superconcept) \in \mathscr{R}'$, c_i is the closest superconcept of c_j.

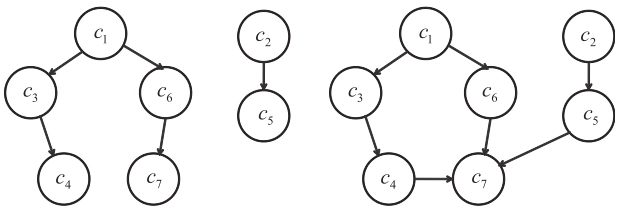
To construct the minimal semantic network, we first sort the concepts $\mathscr{C} = \{c_1, \dots, c_n\}$ according to their sizes in an ascending order. Then, for each concept c_i, we check the superconcept–subconcept relationship from c_{i+1} to c_n and stop checking when we find the first superconcept which will be the closest superconcept of c_i. The algorithm is described in the Algorithm 2.

4.2.2.2. Phase 2. Expanding to compact semantic network. Once the minimal semantic network is built, we expand it to the compact semantic network by generating additional superconcept–subconcept relationships that are not redundant. We first prove by the following Lemma that, given a minimal semantic network $\mathscr{G}$, any additional superconcept–subconcept relationship between c_i and its ancestor c_j will be redundant.

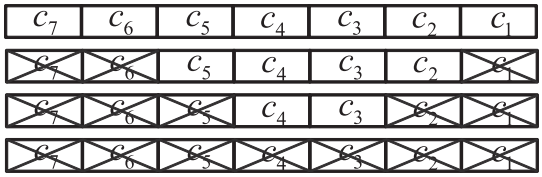
Lemma 2. *Let graph $\mathscr{G}$ be a minimal semantic network. For any concept c_i and its ancestors $c_1, \dots, c_k$ in $\mathscr{G}$, any additional superconcept–subconcept relationships from one of $c_1, \dots, c_k$ to c_i will be redundant.*

It is obvious that any additional path to c_i from its ancestor c_j (i.e., $c_j \to c_i$) is redundant because there exists another path from c_j to c_i (i.e., $c_j \to \cdots \to c_i$) since c_j is an ancestor of c_i. (Note that we defined redundancy by using existence of alternative route consists of other superconcept–subconcept relationship.) From Lemma 2, we can skip examining the superconcept–subconcept relationship from c_i to its ancestors. We illustrate the expansion algorithm by the following example.

Example 1. Fig. 3a–c respectively show an example of minimal semantic network, compact semantic network, and four lists of concepts sorted according to their sizes. From the minimal semantic network, suppose we examine to find another non-redundant superconcept of c_7. Then, we can exclude c_7, c_6, c_1 from



(a) minimal semantic network (b) compact semantic network



(c) Candidate list for expansion

Fig. 3. Example for expanding semantic network.

consideration for additional construction of non-redundant superconcept–subconcept relationship because they are ancestors of c_7 or itself (shown at the second list of Fig. 3c). After that, we test the relationship between c_5 and c_7 and find new superconcept–subconcept relationship from c_5 to c_7. Next, we can exclude c_5 and c_2 because c_2 became an ancestor of c_7 (shown at the third list of Fig. 3c). If we repeat this process until we test all the elements in the list, we will discover entire additional non-redundant superconcepts of c_7. After testing this for all the nodes in the minimal semantic network, we finally get the compact semantic network of Fig. 3b.

Algorithm 2. Compact semantic network generation

Input: a sorted list of concepts C, a document set D, user thresholds θ.
Output: a set of triplet $R = \{(c_i, c_j, Type)\}$.

```
Set R := ∅
// Build minimal semantic network
for i = 1 ··· n − 1 do
  for j = i + 1 ··· n do
    if p(c_j|c_i) > θ_p and p(c_i|c_j) < θ_p then
      R:=R ∪ (c_j, c_i, Superconcept);
      Exit loop;
    end if
  end for
end for
// Expand to compact semantic network
for i = 1 ··· n − 1 do
  L[1:n] = true
  L[Ancestors of c_i] = false
  for j = i + 1 ··· n do
    if L[j] == true and p(c_i|c_j) < θ_p and p(c_j|c_i) > θ_p then
      R:=R ∪ (c_j, c_i, Superconcept);
      L[c_j and ancestors of c_j] = false
    end if
  end for
end for
return R
```

The expansion algorithm is described in the Algorithm 2. Note that, $p_D(c_j|c_i)\left(=\frac{p_D(c_i,c_j)}{p_D(c_i)}\right)$ is computed by using Eqs. (1) and (4). The time complexity of the compact semantic network generation algorithm is $O(n^2h)$ where n is the number of concepts and h is the maximum length of path from the most largest concept. This is because, we have a nest loop and, for each loop, we need spend at most h step to eliminates ancestors from the candidates set.

We prove the correctness of proposing algorithm through theoretical analyses.

Theorem 1. Algorithm 2 *does not generate a redundant relationship.*

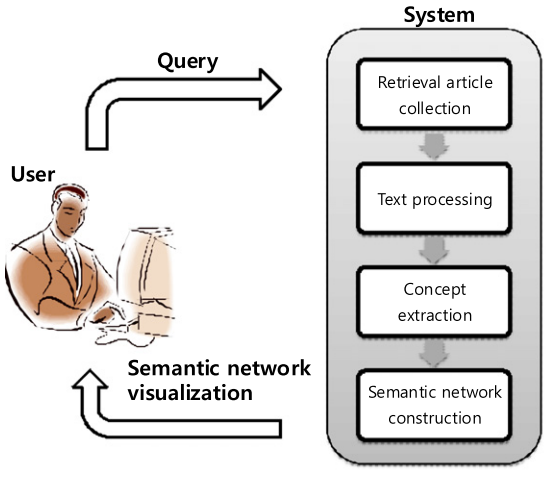
Proof. Suppose, Algorithm 2 generates a redundant relationship and let it $x_i \to x_j$. Since it is redundant relationship, there is another path from x_i to x_j (i.e., $x_i \to \cdots \to x_j$). However, since there is a path from x_i to x_j, x_i is excluded through Algorithm 2 when finding superconcept of x_j. Proof by contradiction. □

Theorem 2. *All possible non-redundant relationship is detected by* Algorithm 2.

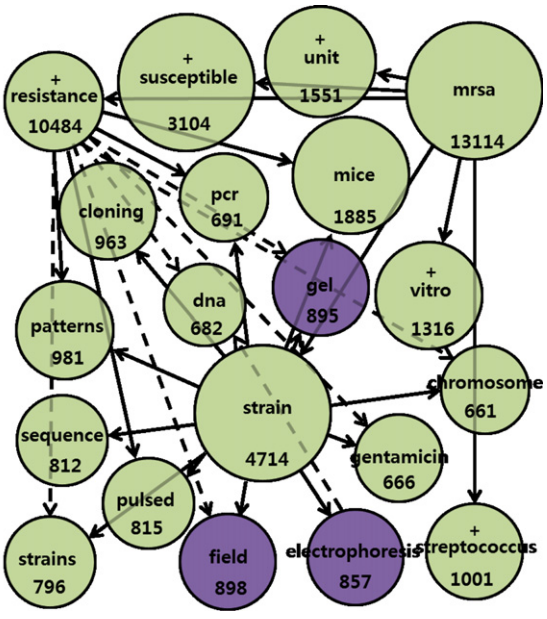
Proof. Suppose, there is a not-detected non-redundant relationship and let it $x_i \to x_j$. Since $x_i \to x_j$, $|x_i| > |x_j|$. And also, since $x_i \to x_j$ is not a redundant relationship, there is no another path from x_i to x_j which implies that x_i will not be excluded while finding superconcept of x_j. Therefore, Algorithm 2 will test the relationship between x_i and x_j and will find the relationship. Proof by contradiction. □

5. Application: RefMed

We evaluate the COMPACT (Algorithm 2) against POST (Algorithm 1) that removes redundancy after construction. For evaluation, we implemented the methods in RefMed, an interactive search engine for PubMed. In this section, we briefly introduce RefMed, and visualization rules that we used to present semantic network.



(a) Framework



(b) Example of semantic network for "mrsa"

Fig. 4. Framework of our system and example semantic network results using a PubMed query "mrsa".

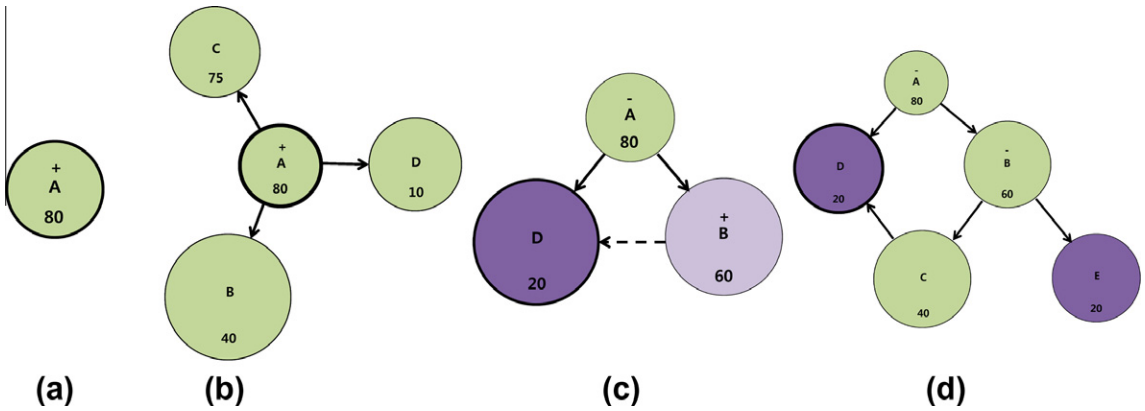


Fig. 5. Example of node expansion, *related* and multiple superconcepts.

5.1. RefMed system overview

Fig. 4a shows the framework of our system. If user enters a keyword, the system (1) first finds exactly matching documents as done in the original PubMed search engine, (2) preprocesses the result documents by running stemming and stopwords, (3) extracts the concepts, (4) constructs semantic network of concepts by extracting the relationship between concepts, and (5) visualizes them by drawing an interactive graph.

Fig. 4b shows an example of the semantic network drawn in our system as a result of query "mrsa". The semantic network contains total 123 concepts including the query concept, 148 superconcept–subconcept relationships, and three *related* relationships. The size of semantic network is adjustable by the concept thresholds. In the figure, the query node has six subconcept concepts, and one of the subconcept concept, "strain", contains 13 subconcept concepts. The concept "gel" is a *related* concept of "electrophoresis". The detailed description of the semantic network structure is provided in Section 5.2.

5.2. Interactive visualization

Visualizing the semantic network is not trivial due to its large size and complex structure. RefMed interactively visualizes the resulting semantic network to let users intuitively explore the semantic network. RefMed first expands only the subconcept concepts from the query concept. If the user wants to explore any subconcept concept more deeply, then the node of the concept will be expanded. The followings summarize the basic rules for RefMed visualization.

- Each node represents a concept: the size of node reflects the importance of the concept measured by its frequency entropy computed by using Eq. (7), and the number in the node is the frequency of the concept in the result set.

$$entropy(x_i) = -p_D(x_i)\log(p_D(x_i)) \tag{7}$$

- '+' indicates there are hidden subconcept concepts of the node which will be expanded when it is clicked.
- The directed line indicates the superconcept–subconcept relationship between the nodes.

Fig. 5 shows a simple example explaining the above rules. The size of result set is assumed to be 100. Fig. 5a shows that the root concept 'A' occurs in 80 articles and there are hidden subconcept concepts. When the node is clicked, its subconcept concepts are displayed as shown in Fig. 5b, in which the directed line between 'A' and 'B' indicates 'A' is a superconcept of 'B'. Note that the subconcept concepts nodes have different sizes depending on their importance scores.

The followings summarize the visualization rules for *related* set and hidden superconcept relationship.

- Related concept nodes will be colored differently when the user locates the mouse point over one of the nodes.
- If one of the *related* concept has a shaded color, it indicates that a subconcept of the shade-colored node is a *related* concept of the other colored nodes.
- The dotted line from c_i to c_j indicates that one of subconcept of c_i has a superconcept–subconcept relationship with c_j.

Fig. 5c and d shows an example explaining the visualization for *related* relationship and hidden superconcept relationship. In Fig. 5c, the nodes 'B' and 'D' are colored when the user locates the mouse point over one of nodes. Note that the shaded color of node 'B' indicates that a subconcept of 'B' is a *related* concept of 'D'. There is also a dotted line from 'B' to 'D' which indicates that a subconcept of 'B' is a superconcept of 'D'. Node 'B' will display its hidden subconcept concepts when it is clicked, which is shown in Fig. 5d. In the figure, we see that nodes 'D' and 'E' are *related* concepts of each other and node 'C' is a superconcept of 'D'.

6. Experimental results

6.1. Implementation

The dynamic construction of semantic network is implemented in the PHP (v5.1.6) and the Adobe Flash CS4. The algorithm is implemented using the PHP and the visualization is implemented using the Flash. The implementation is integrated into our RefMed. The RefMed is implemented using MySQL (v5.1) and C language. We store the entire PubMed data in RefMed. Our system is accessible at "http://dm.postech.ac.kr/refmed/". The system currently

runs on a server of quadcore 2.87 GHz with 46 M memory, thus cannot support many simultaneous accesses at this point. In our system, stemming and stop words removal are processed for each articles, and each article is represented in form of a tf/idf vector. Once a keyword query is submitted to the system, corresponding documents are retrieved, and a semantic network is constructed from the documents.

6.2. Semantic network construction time evaluation

We evaluate the time of semantic network construction on queries having results of varying sizes. Constructing a network includes counting co-occurrence, frequency and inverted document frequency for each term from retrieved documents. Since this part is common for all methods, we do not include the time for this process in comparison. The running time of post-processing method, POST, includes the time for constructing the whole network (detecting relationship part in Algorithm 1) and removing redundant relationships (removing redundant relationship part in Algorithm 1). Similarly, the running time of the proposed method, COMPACT, includes the time for building minimal semantic network and compact semantic network. Table 1 reports the construction time. Our method took less than 0.3 s for all queries. On the other hand, in worst case, POST took more than 1.5 s which is seven times longer than our method.

Table 1
Network construction time.

Query	# Of doc.	# Of concepts	POST (s)	COMPACT (s)
Mrsa	14,345	123	1.0931	**0.0638**
Hemophilia	20,039	55	0.0688	**0.0417**
Methyltransferase	28,153	99	0.3606	**0.1367**
Melanoma	76,274	146	1.0930	**0.1665**
Malignant	239,278	171	1.5557	**0.2269**
Cytokine	469,843	51	**0.1614**	0.2204

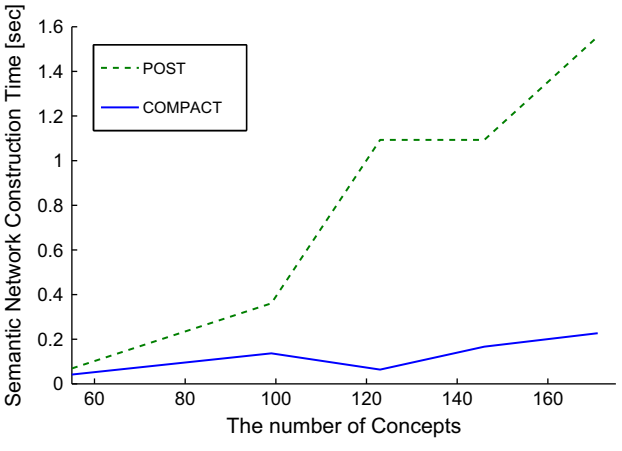


Fig. 6. The change of construction time as the number of concept increases.

Fig. 6 illustrates the construction time on an increasing number of concepts. The construction time of POST rapidly increases as the number of concepts increases. On the other hand, our method produces stable construction time. This result does not seem to comply the time complexity analysis results since the time complexities of POST and COMPACT are $O(n^4)$ and $O(n^2h)$ respectively. However, the time complexity is written in big O bound and it can vary according to the topology of semantic network. From an investigation on the obtained semantic networks, we found that the semantic networks from real data are far from complete in terms of connections between nodes. Thus, the number of relationships hardly becomes n^2. As a result, the time difference is not as large as we expected.

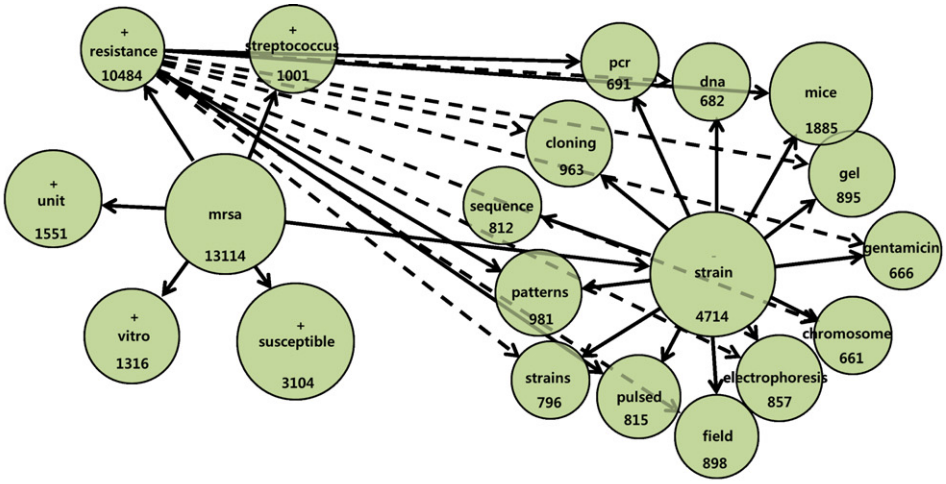
6.3. Semantic network qualitative evaluation

We qualitatively evaluate the effectiveness of our system. We also evaluate our system in qualitative way. Fig. 7a is a typical result for the keyword "mrsa". In this figure, each node represent the concept in the semantic network, and the number on each node indicates the occurrence of concept in the search results. In this example, strain occurs in 4147 articles in the search results. In the case of resistance, the search term occurs in 10,484 articles. The size of keyword resistance is smaller than that of strain because resistance occurs too often in the list. We processed qualitative evaluation by adding the two terms to the query independently, and comparing how much useful information is generated from each extension. We pick two new nontrivial concepts, "resistance" and "strain", detected by our system.

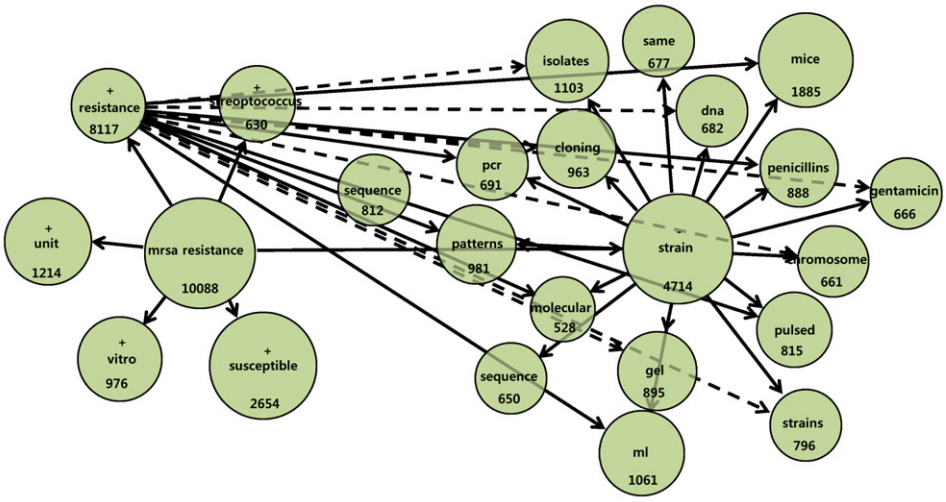
Fig. 7b is a result of the query keyword "mrsa resistance". Note that, the term "resistance" is not considered as good concept in semantic network for mrsa. The semantic network generated from the extended keyword "mrsa resistance" shows similar result compared to that of original keyword "mrsa". There is no many newly added concepts, and even the size of concepts are not changed much. In addition, structural arrangement of concepts are also similar to that of "mrsa" We can conclude that the keyword "resistance" does not carry any additional information, as we expected.

We also analyzed the search result by using the keyword "mrsa strain" again (Fig. 7c). Clearly, the semantic network result of "mrsa strain" is different from that of "mrsa". New keywords such as "oxacillin", "vancomycin", "ciprofloxacin" appear in the semantic network. In addition, structural arrangement is quite different from that of "mrsa". From this comparison, we can conclude that the goodness of concept is good for extending queries.

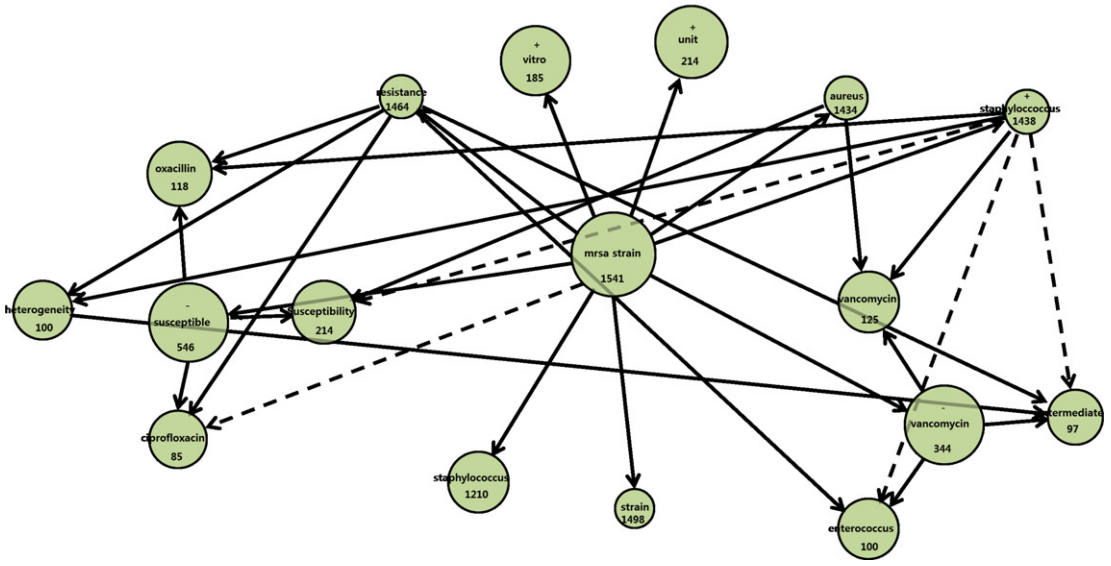
During exploration, we learned that "mrsa" has relationship with several terms "resistance", "streptococcus", "susceptible", "strain", "oxacillin", "vancomycin". Actually, the keyword "mrsa" stands for methicilin resistance staphylococcus aureus which is a type of bacteria that cannot be killed by anti-biotics. More specific, "mrsa" is a sort of strain (strain means genetic variants or subtypes of a micro-organism), which has resistance to some beta-lactam antibiotics, which include but not limited to the antibiotics such as oxacillin, vancomycin, ciprofloxacin. The specific type of mrsa is determined according to the antibiotics. From this investigation, we can conclude that semantic network in our system is good for capturing important terms for the query keyword.



(a) The result of the keyword 'mrsa'



(b) The result of the keyword 'mrsa resistance'



(c) The result of the keyword 'mrsa strain'

Fig. 7. Query results for qualitative evaluation.

7. Conclusions

In this paper, we proposed an efficient construction method for semantic network. Traditional semantic network methods generate many relationships, but most of them do not show any useful information. Thus, those redundant relationship should be removed to clearly present the underlying important relationship among concepts. However, removing these redundant relationship take much time as the number of concept increases. This paper proposed a novel efficient method that constructs semantic network having no redundant relationship. We implemented our method in RefMed to evaluate the efficiency. Semantic network construction time of the POST method increases rapidly as the number of concepts increases. In contrast, COMPACT produces stable construction time and do not rapidly increase.

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